

COBOURG, ON, Canada

WMO: 714310

Lat: 43.9561N

Lon: 78.1511W

Elev: 255

StdP: 14.56

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.8	(c) 2.8	(d) -18.5	(e) 2.0	(f) -1.2	(g) -10.9	(h) 3.1	(i) 4.9	(j) 32.0	(k) 28.6	(l) 29.1	(m) 29.2	(n) 7.6	(o) 320	(p) 0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.4	78.6	70.1	76.6	69.7	74.9	68.9	73.1	76.4	71.6	75.1	70.1	73.6	6.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	119.3	75.4	70.3	112.6	74.1	68.8	106.8	72.8	36.9	76.4	35.6	75.1	34.3	73.6	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.4	21.1	18.0	-8.8	84.0	5.5	3.8	-12.8	86.8	-16.0	89.0	-19.1	91.1	-23.1	93.9
			-9.5	75.3	5.1	1.8	-13.2	76.6	-16.2	77.6	-19.1	78.6	-22.8	79.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	22.8	24.5	31.6	42.6	52.5	61.6	67.4	67.4	61.5	49.6	38.9	29.5
	DBStd	17.46	11.51	9.27	8.73	6.30	6.16	5.41	4.75	4.52	6.04	7.03	7.46	9.09
	HDD50	3475	844	715	570	233	44	0	0	0	2	96	335	636
	HDD65	7226	1309	1135	1034	673	390	130	30	27	136	479	782	1101
	CDD50	1993	0	0	1	10	122	348	538	540	348	83	3	0
	CDD65	267	0	0	0	0	2	28	103	102	32	0	0	0
	CDH74	600	0	0	0	0	6	41	246	247	60	0	0	0
	CDH80	41	0	0	0	0	0	2	25	14	0	0	0	0
(14) Wind	WSAvg	7.3	9.5	9.1	8.5	8.1	6.5	5.3	5.0	5.4	6.1	7.3	8.5	9.1
(15) Precipitation	PrecAvg	34.7	2.6	2.1	2.4	3.1	3.0	3.0	3.2	3.0	3.2	3.3	2.8	2.8
	PrecMax	40.9	5.0	4.3	4.9	5.0	5.6	6.1	6.5	4.9	6.1	6.6	4.4	5.8
	PrecMin	29.6	1.3	0.9	1.1	1.1	0.9	0.7	1.4	1.1	1.8	1.1	1.2	1.7
	PrecStd	3.5	1.0	0.8	0.9	1.2	1.3	1.5	1.4	0.9	1.1	1.4	0.9	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.2	45.6	53.4	62.1	72.7	78.2	83.3	81.8	78.3	68.7	56.4	49.7
		MCWB	45.7	41.9	46.1	50.7	63.1	67.0	68.2	69.9	71.8	64.4	52.4	46.9
	2%	DB	42.7	41.3	48.7	56.7	67.2	74.5	78.8	78.6	75.7	65.9	53.9	46.4
		MCWB	41.1	38.4	43.7	47.8	60.1	66.4	69.5	71.0	69.8	61.7	50.3	44.3
	5%	DB	40.0	38.7	45.3	53.6	64.2	71.8	76.5	76.5	73.5	63.4	51.7	43.8
		MCWB	37.9	35.8	41.0	46.4	57.5	65.3	69.2	70.5	68.1	59.2	48.6	41.4
	10%	DB	37.5	36.8	42.3	51.0	61.0	69.5	74.6	74.9	71.3	60.8	49.5	41.3
		MCWB	35.4	34.1	38.2	45.1	55.1	63.9	68.4	69.4	66.3	56.5	46.5	39.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.1	42.4	49.1	54.2	65.8	71.2	74.5	75.4	73.5	66.5	53.9	47.7
		MCDB	46.8	44.4	51.5	59.5	70.6	74.9	77.9	77.8	76.8	67.4	55.6	49.2
	2%	WB	41.4	38.7	44.8	50.6	61.5	68.5	72.6	73.6	71.2	62.4	51.2	44.5
		MCDB	42.4	40.7	47.2	54.7	65.6	72.2	76.1	76.6	74.7	65.3	53.0	46.2
	5%	WB	38.1	36.3	41.7	48.0	58.8	66.6	71.0	72.1	69.2	60.0	49.1	42.0
		MCDB	39.6	38.4	44.7	51.8	62.9	70.1	74.6	75.3	72.4	62.7	51.3	43.6
	10%	WB	35.5	34.4	38.9	45.9	56.2	64.7	69.6	70.6	67.3	57.4	47.1	39.1
		MCDB	37.3	36.5	41.8	49.9	60.1	68.5	73.2	73.9	70.6	60.2	49.3	41.0
(35) Mean Daily Temperature Range	5% DB	MDBR	13.0	12.9	12.0	12.3	12.7	12.6	12.2	12.4	13.1	11.8	11.1	11.0
		MCDBR	11.1	11.7	13.3	16.0	16.2	15.0	13.9	13.2	13.3	13.2	12.5	11.3
	5% WB	MCWBR	11.3	10.8	10.5	11.0	10.4	9.0	8.6	9.6	10.2	11.0	11.4	10.9
		MCDWR	11.4	11.7	12.5	14.2	14.5	13.0	11.6	11.2	11.9	12.2	11.5	10.8
(40) Clear-Sky Solar Irradiance	taub		0.306	0.311	0.325	0.363	0.389	0.424	0.421	0.411	0.373	0.358	0.314	0.290
		taud	2.377	2.366	2.424	2.372	2.349	2.282	2.303	2.320	2.409	2.430	2.506	2.483
	Ebn at Noon		259	279	291	286	280	269	269	267	270	259	256	257
		Edh at Noon	25	30	32	36	39	42	40	38	32	28	22	20
(44) All-Sky Solar Radiation	RadAvg		414	684	1099	1420	1736	1856	1886	1657	1321	781	481	335
	RadStd		41	61	103	151	180	148	142	85	109	70	61	43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page